

Development of a Compliant Robotic Gripper

Self-Locking Mechanism and E-textiles

SYSMER Track | 2 Year Engineering Internship | MicroNano Systems (WSN)

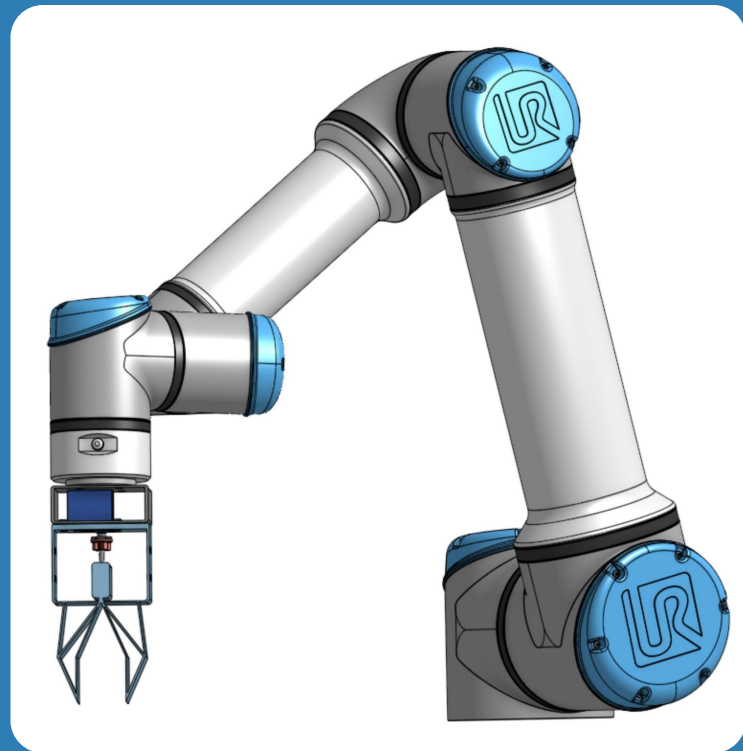
From 11 May to 28 August 2026

Supervisor

Matteo Menolotto

Teacher

Cédric Anthierens



Julien Dubuc



Problem & Goal



<https://blog.robotiq.com/the-simple-guide-to-choosing-between-so-many-grippers>

Current System: Classic Rigid Grippers

Energy-intensive

No sensors

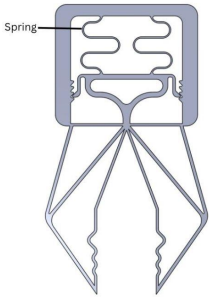
Risk to fragile objects



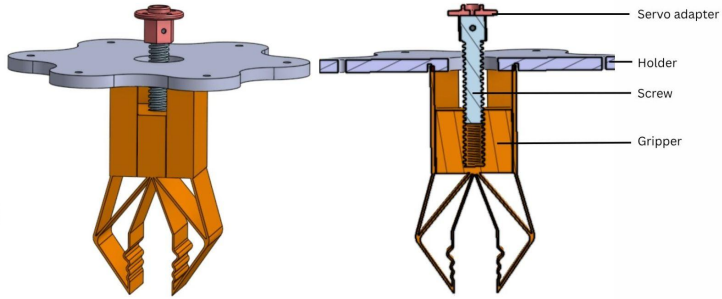
Alternative: Underactuated Compliant Gripper

Goal: Grasp objects, hold them (0W), and release them in a controlled manner to ensure the safety of fragile objects. Integrate e-textiles to provide tactile feedback.

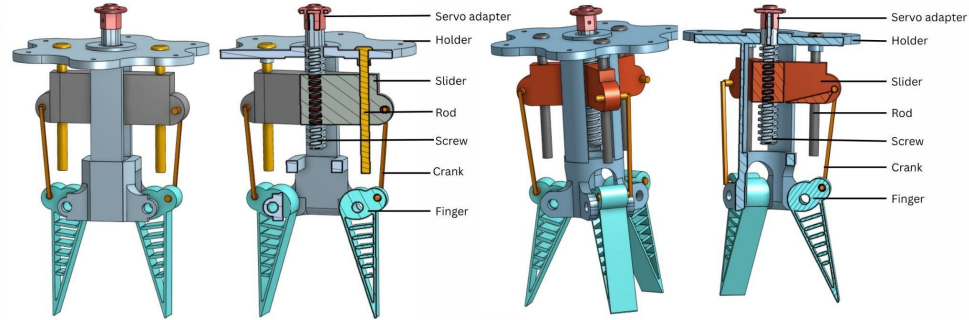
CAD Prototypes



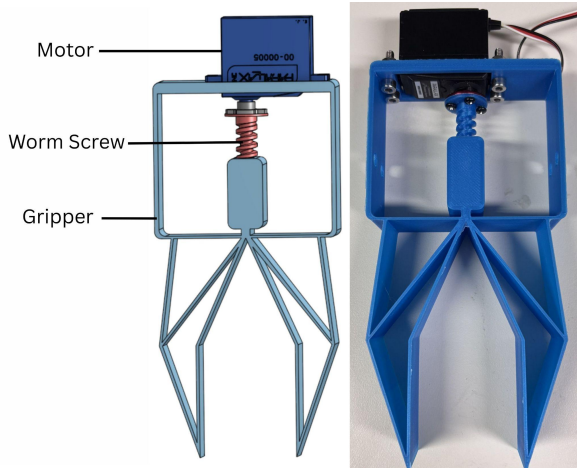
Inspiration



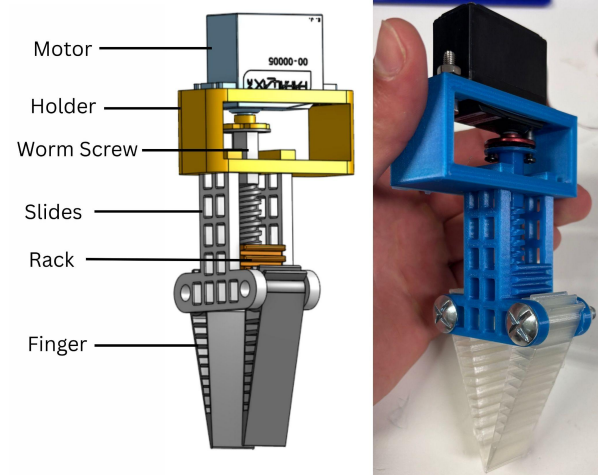
Design 1: Initial Compliant Gripper



Design 2: Slider-Crank Gripper

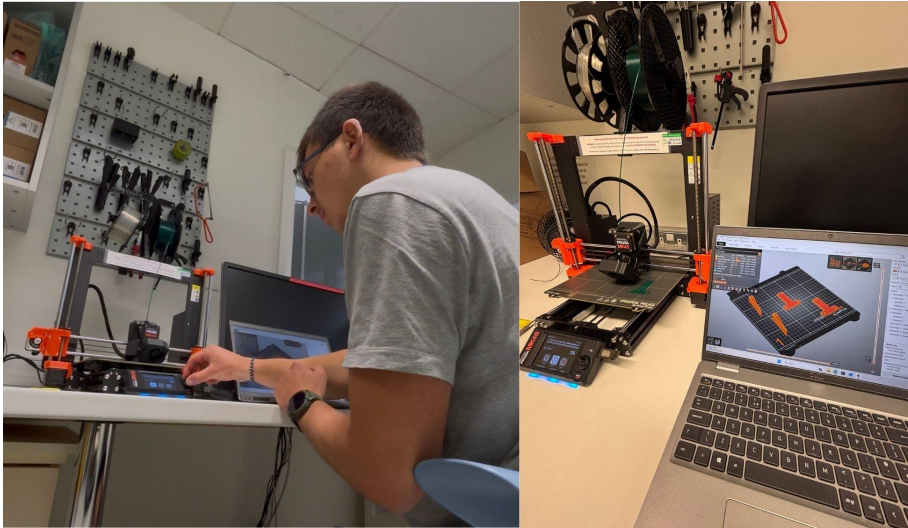


Design 1.1: Compliant Gripper (2 Parts)

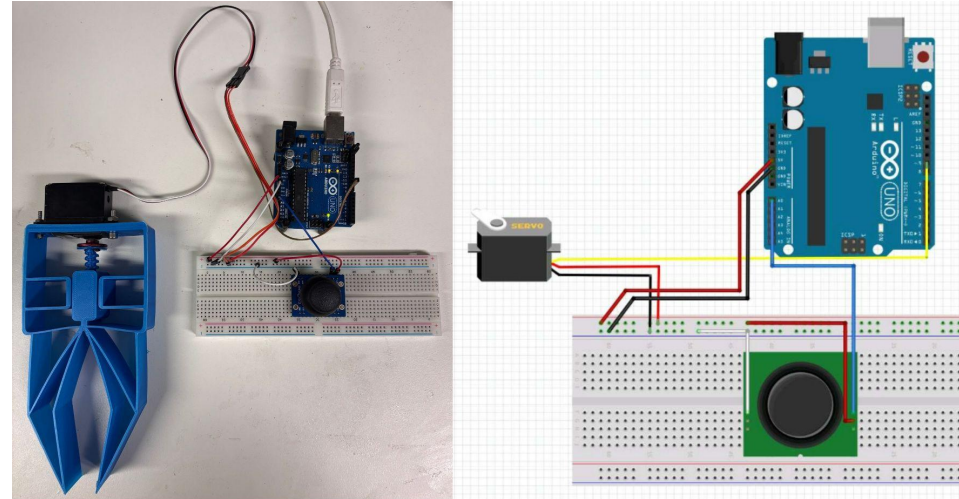


Design 3: Rack and Pinion Gripper

Fabrication & Iteration

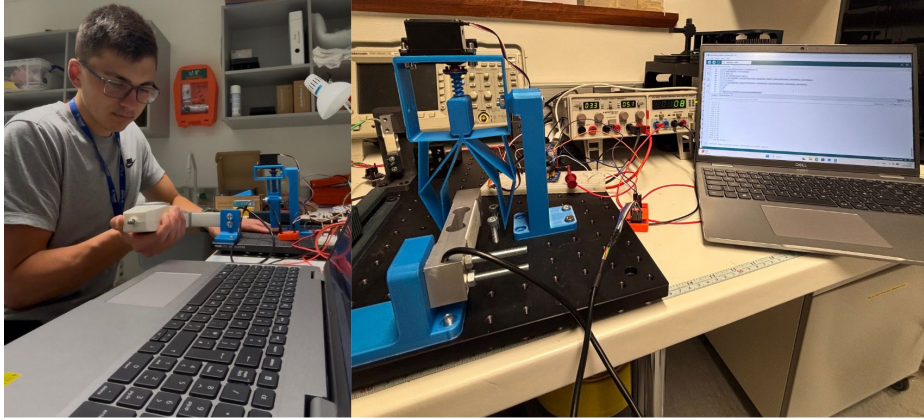


3D Printing: Prusa MK4S, PrusaSlicer Software

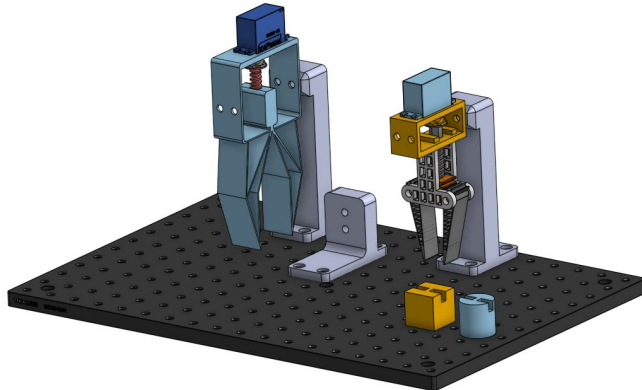


Basic Electronics Setup: Arduino UNO, Joystick & Servomotor

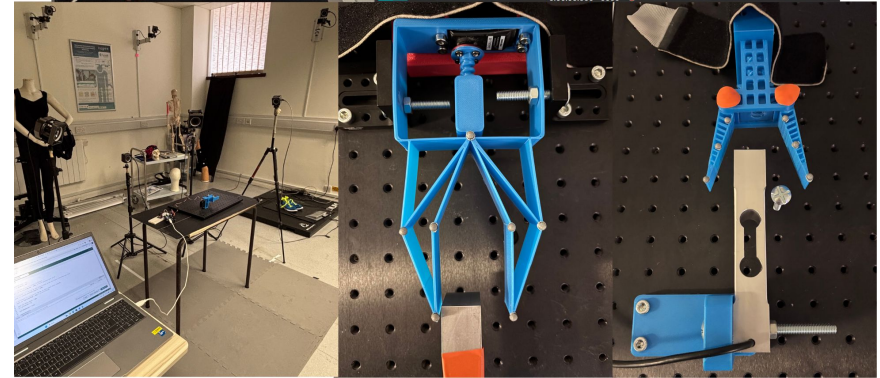
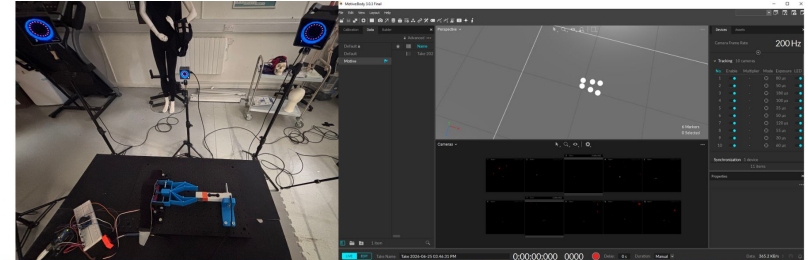
Protocol Assessment & Setup



Test 2 Setup: Calibration & Measurement



Testing Table: Holders, Objects



Test 1 Setup: Motion Capture System, Grippers & Load Cell

Test 1
Kinematics &
Stiffness

Test 2
Force, Power &
Efficacy

Test 3
Zero-Energy
Hold (Passive
Locking)

Test 4
Durability &
DFMA Evaluation

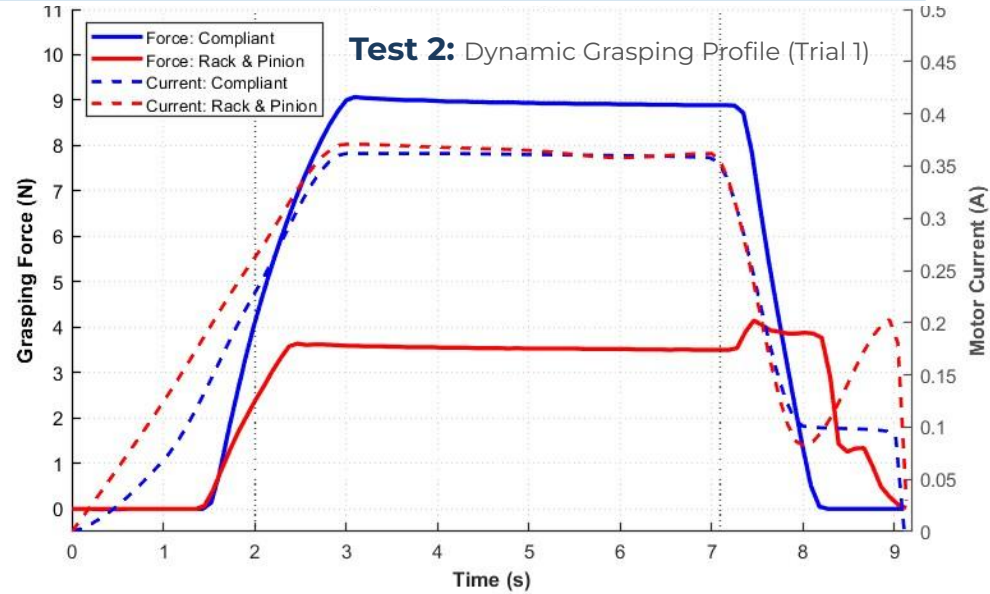
Key Results: Test 1 & 2

Metric	Compliant Gripper	R&P Gripper
Max Opening (mm)	73.60 ± 1.39	144.12 ± 1.49
Min Closing (mm)	0	0
Opening Time (s)	8.0	4.3

Test 1: Kinematic Specifications (3 trials)

Metric (3 trials)	Compliant Gripper	R&P Gripper
Max Horizontal Force	17.60 ± 2.56 N	6.93 ± 0.25 N
Force Ratio	2.54	

Test 1: Max Horizontal Force

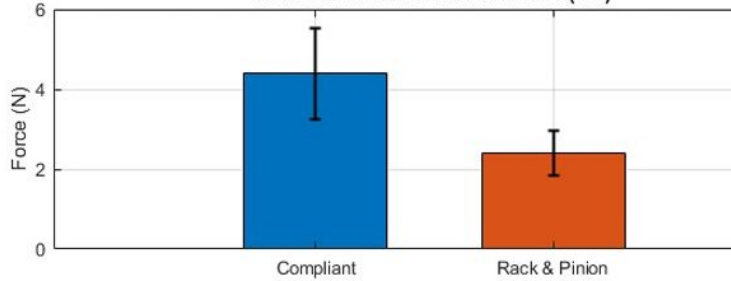


Metric (5 trials)	Compliant Gripper	R&P Gripper
Max Vertical Force (N)	9.12 ± 0.29	3.50 ± 0.27
Peak Current (A)	0.365 ± 0.002	0.372 ± 0.002
Holding Current (A)	0.362 ± 0.002	0.368 ± 0.004
Holding Power (W)	1.81 ± 0.01	1.84 ± 0.02
Efficacy (N/W)	5.04 ± 0.17	1.91 ± 0.15

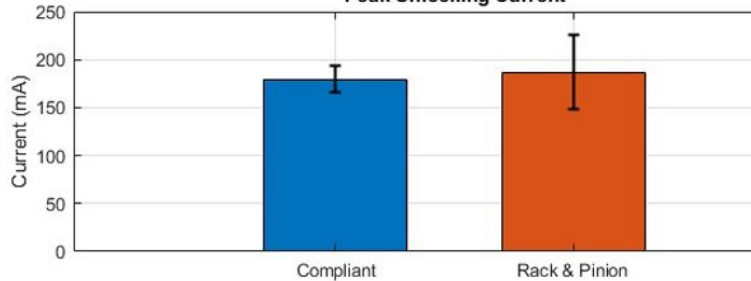
Test 2: Efficacy & Electrical Metrics

Key Results: Test 3 & 4

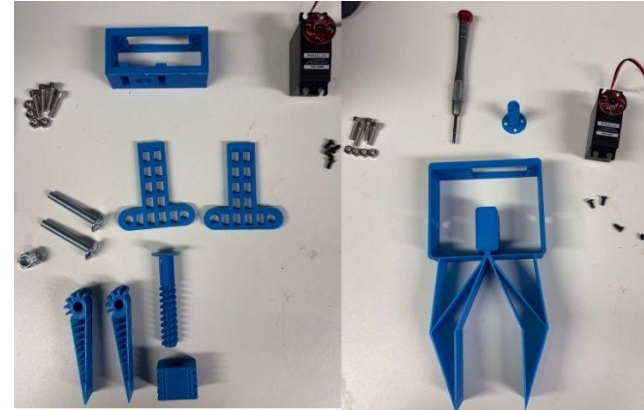
Maximum Passive Pull-out Force (0W)



Peak Unlocking Current



Test 3: 5 trials

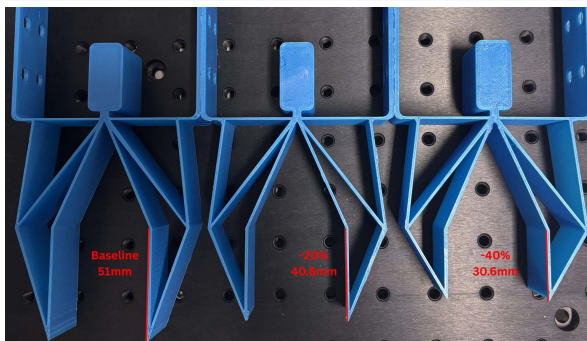


Disassembled Gripper: R&P and Compliant

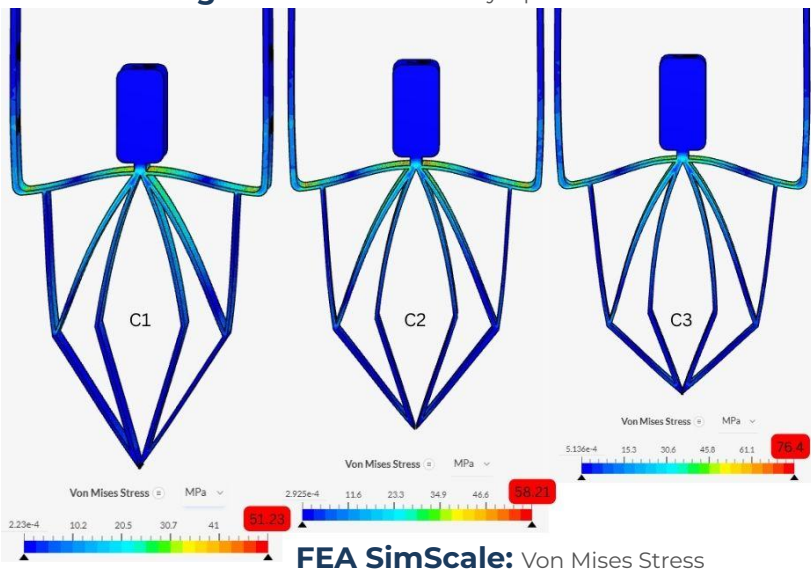
Metric	Rack & Pinion Gripper	Compliant Gripper
Number of Parts	7	2
Printing Time	3h15m	2h12m
PLA Used (g)	52.91	54.67
Printing Cost (€)	1.48	1.53
Mass (g)	68.98 (7 pieces+2*M6x40+nuts)	50.69 (2 pieces)
Materials	2*M6x40+nuts 4*M4x16+nuts 4*servo screws	4*M4x16+nuts 4*servo screws
Assembly Time	5m 31s	2m 47s

Test 4: DFMA Evaluation

Key Results: Compliant Gripper Optimization



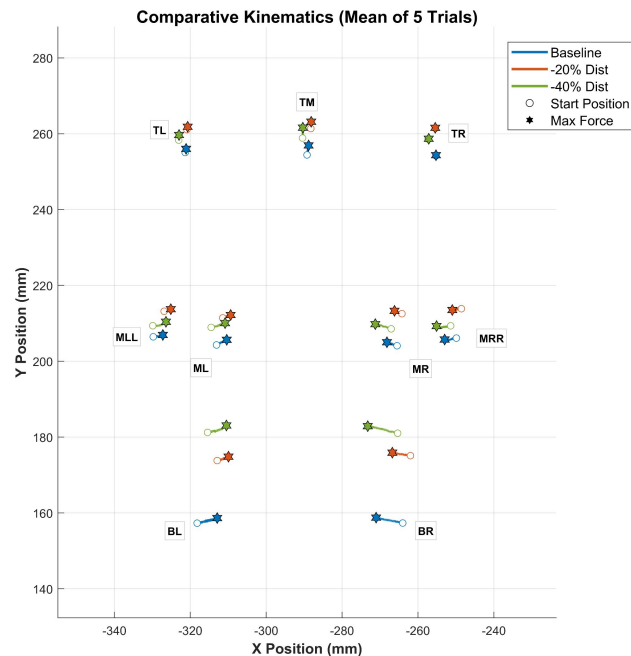
Design Variants: Geometry Optimization



FEA SimScale: Von Mises Stress

Test A
Max Force,
Current, Power
and Efficacy

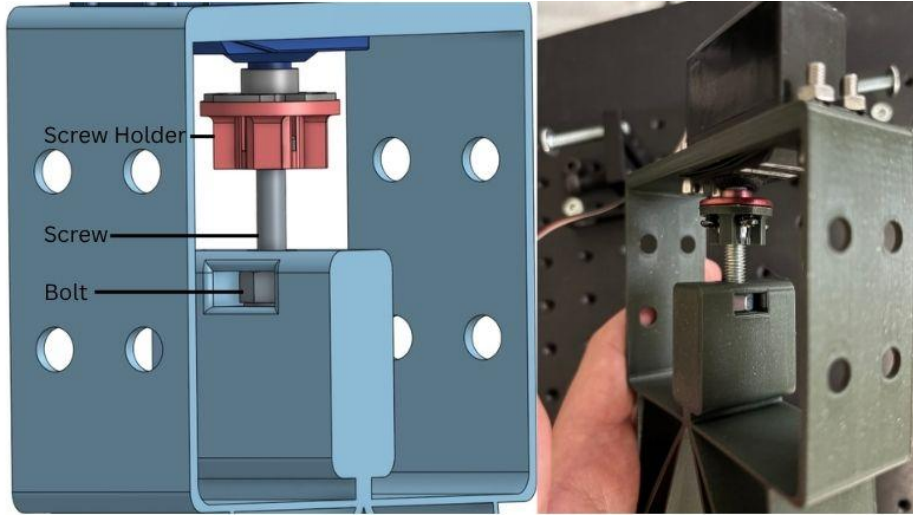
Test B
Deformation
Study



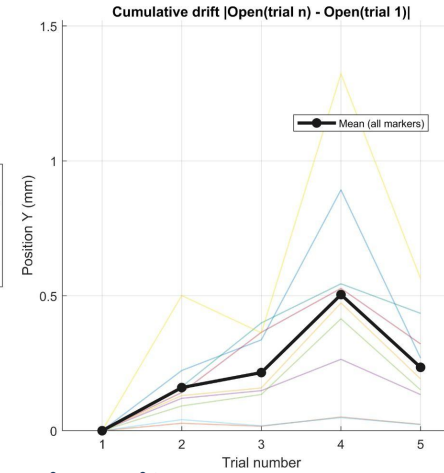
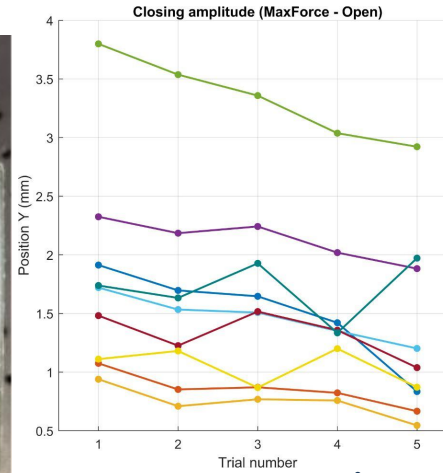
Metric (5 trials)	Baseline	-20%	-40%
Max Force (N)	21.49 ± 0.38	26.42 ± 0.84	25.28 ± 0.95
Peak Current (A)	0.458 ± 0.002	0.453 ± 0.004	0.455 ± 0.003
Peak Power (W)	2.353 ± 0.009	2.324 ± 0.020	2.338 ± 0.012
Actuation Efficacy (N/W)	9.13 ± 0.15	11.37 ± 0.29	10.82 ± 0.41
Mass (2 Parts) (g)	50.69	49.45	48.32
Force-to-Mass Ratio (N/kg)	423.95	534.28	523.18

Test A: Comparison Results

Key Results: Metal Transmission Optimization



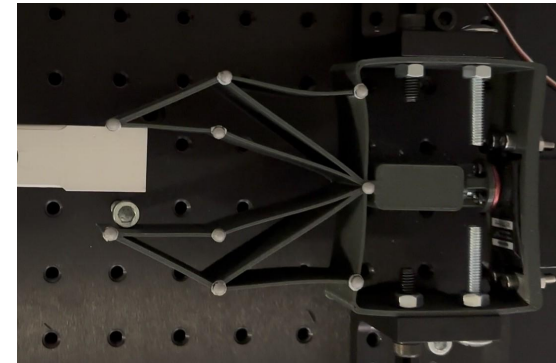
CAD Modification: Metal Transmission Gripper



Hysteresis and Cumulative Drift: 5 trials

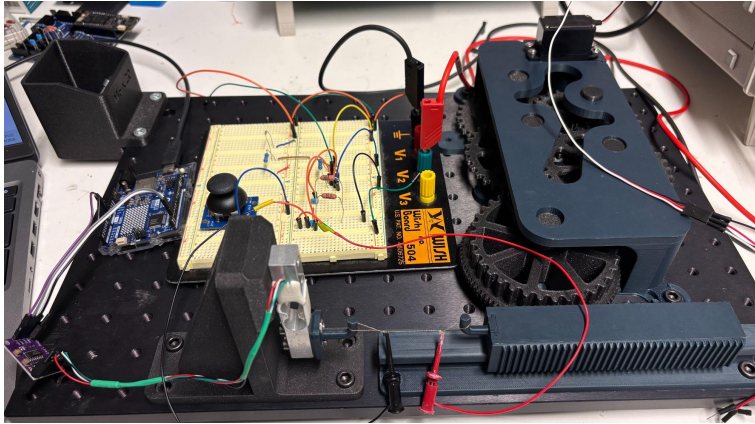
Metric (5 trials)	C2 (PLA Transmission)	C2 (Metal Transmission)
Max Force (N)	26.42 ± 0.84	31.24 ± 2.62
Peak Current (A)	0.453 ± 0.004	0.430 ± 0.009
Peak Power (W)	2.324 ± 0.020	2.216 ± 0.047
Actuation Efficacy (N/W)	11.37 ± 0.29	14.12 ± 1.38

Test A: Comparison Results

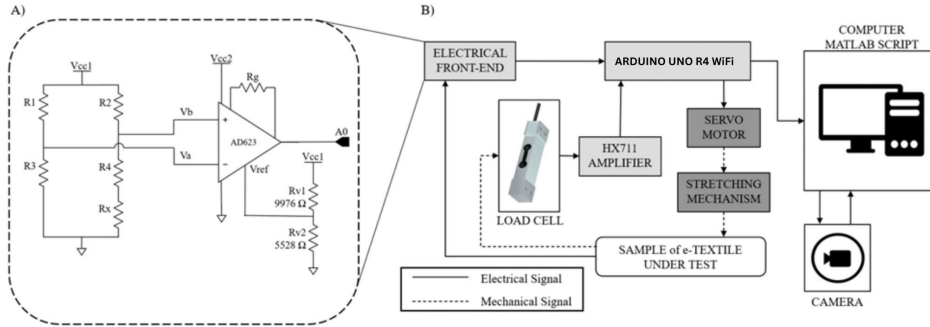


Limit: Bend Inwards at Max Torque

E-textiles & Integration



E-textiles Characterization Setup

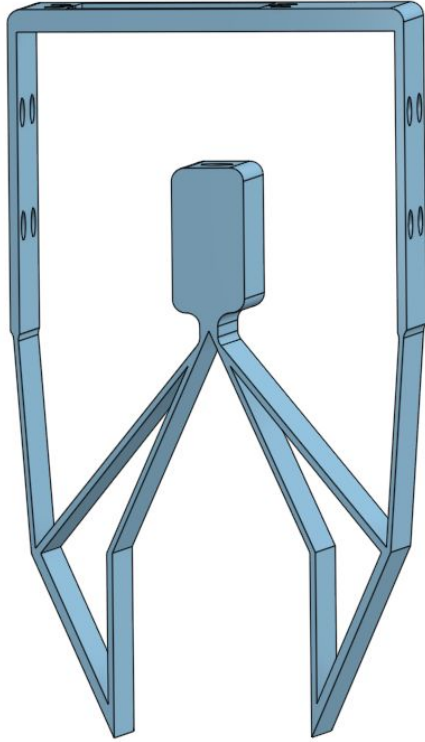


Brother Innov-is NV2700 Embroidery Machine

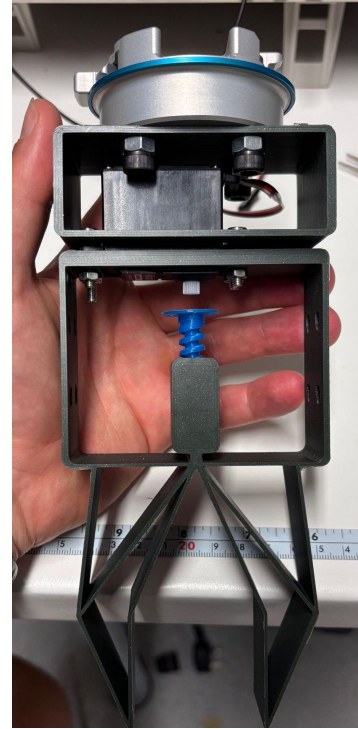


Embroidered Fabric on Foam Pads

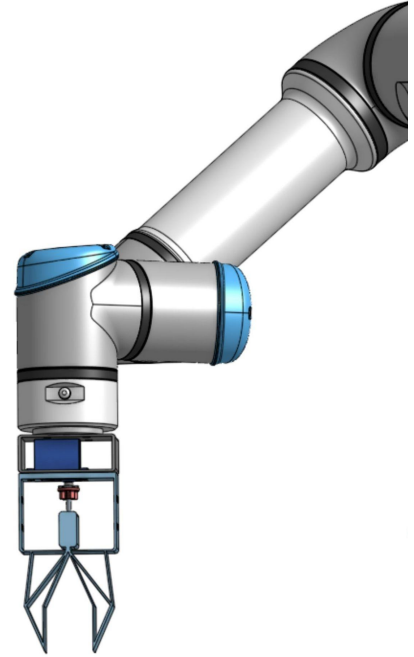
Next steps



New Design



Gripper/Robotic Arm Design



Thank you
Questions ?

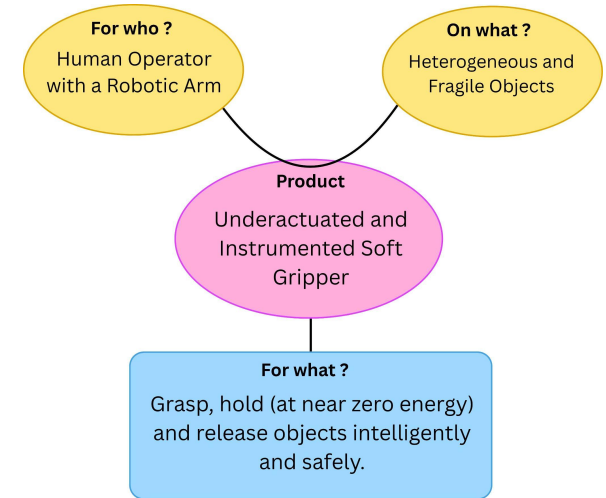
Appendix



Labs

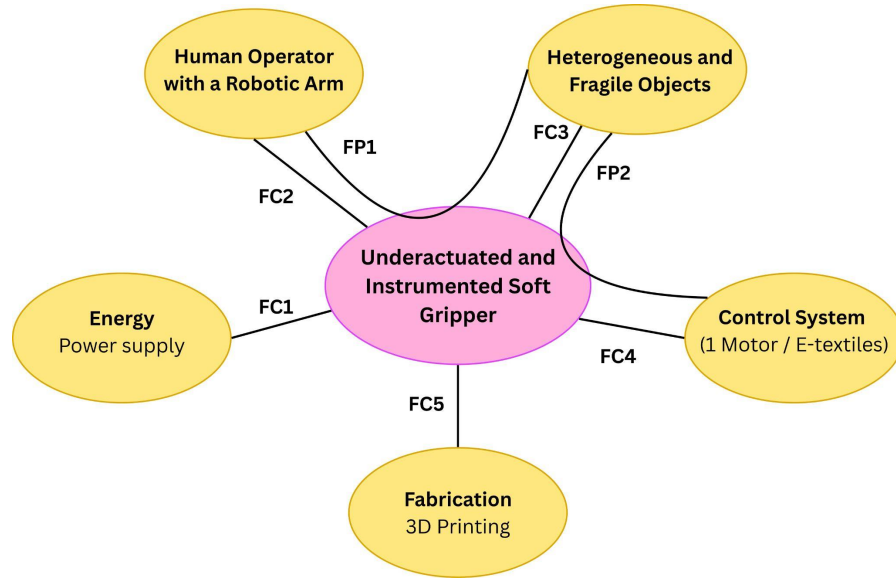


Tyndall National Institute



Need Diagram

Appendix

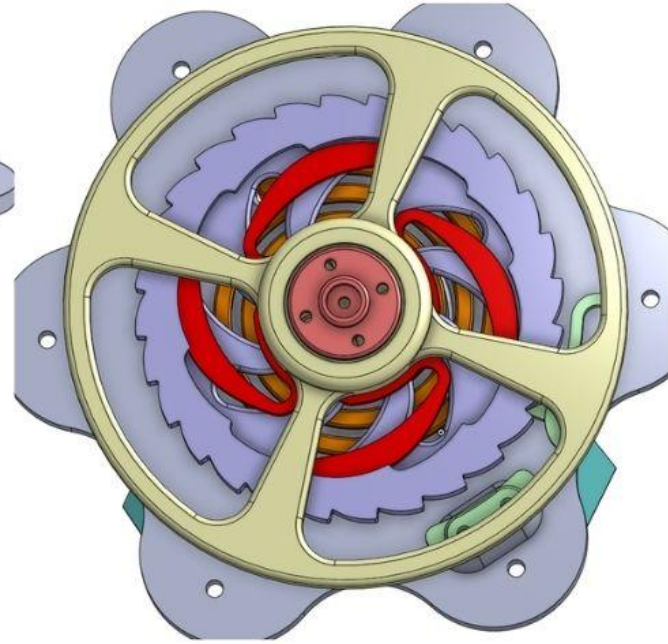
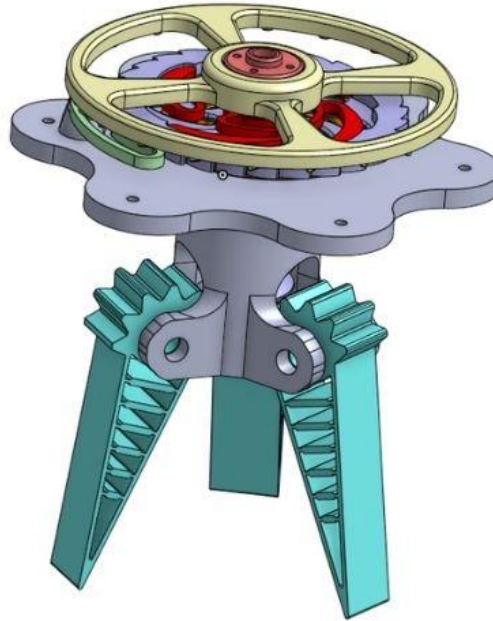
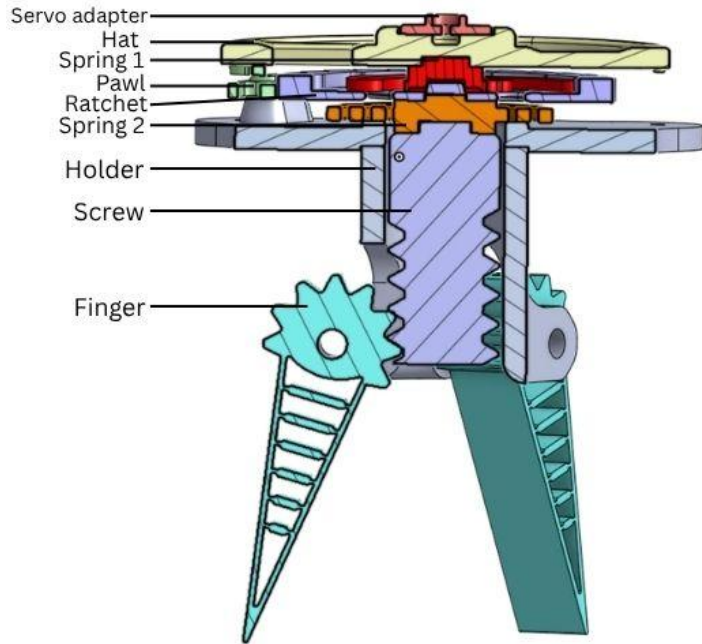


Octopus Diagram

Ref	Function	Assessment Criterion	Expected Level
FP1	Grasp & release	Min opening range	0-70 mm
		Passive holding force (0W)	≥ 3 N
		Opening time	≤ 10 s
FP2	Force transmission	Sensor precision	$\pm \dots$ N
FC1	Energy efficiency	Holding power	0 W
		Actuation efficacy	≥ 5 N/W
FC2	Operator safety	Max contact force	≤ 50 N
FC3	Object integrity	Surface damage	None
FC4	Control architecture	Actuation / Sensors	1 motor / E-textiles
FC5	Low-cost fabrication	Production / Cost	3D Print / ≤ 5 €
		Parts / Assembly time	minimally / < 8 min
FC6	Durability	Lifespan / Force retention	> 500 cycles / $> 70\%$

Table 4.1: Product Requirements Document

Appendix



Previous Lab Design

Appendix

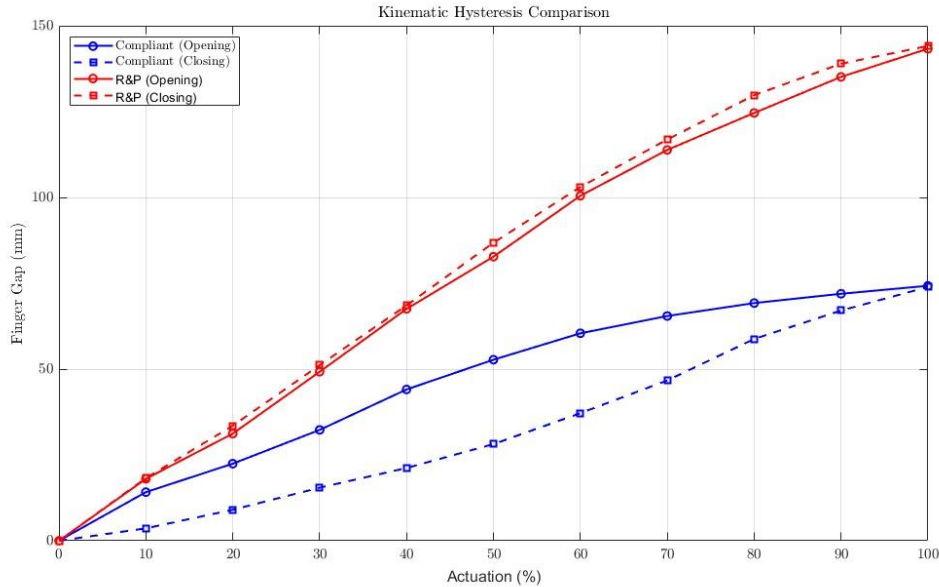
The self-locking feature of a worm screw depends on its lead angle λ and the friction μ between the parts. To prevent the screw from turning backward on its own, we need to have: $\tan(\lambda) < \mu$. I designed it with a screw pitch $p = 4\text{mm}$ and a mean diameter $d_m \approx 8.5\text{mm}$:

$$\lambda = \arctan\left(\frac{p}{\pi \cdot d_m}\right) = \arctan\left(\frac{4}{\pi \cdot 8.5}\right) \approx 8.5^\circ$$

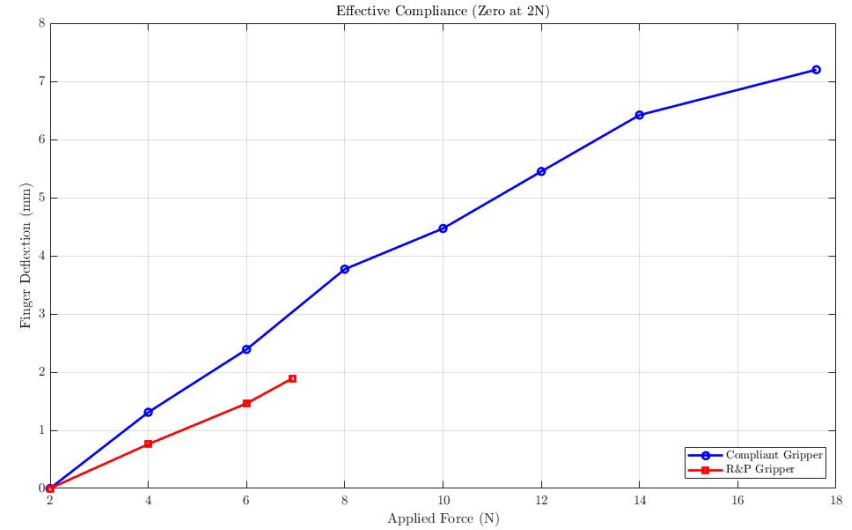
So $\tan(8.5^\circ) \approx 0.15$ is lower than the PLA friction coefficient ($\mu \approx 0.3$). So the mechanism naturally locks itself by design.

Self-Locking Principle

Appendix



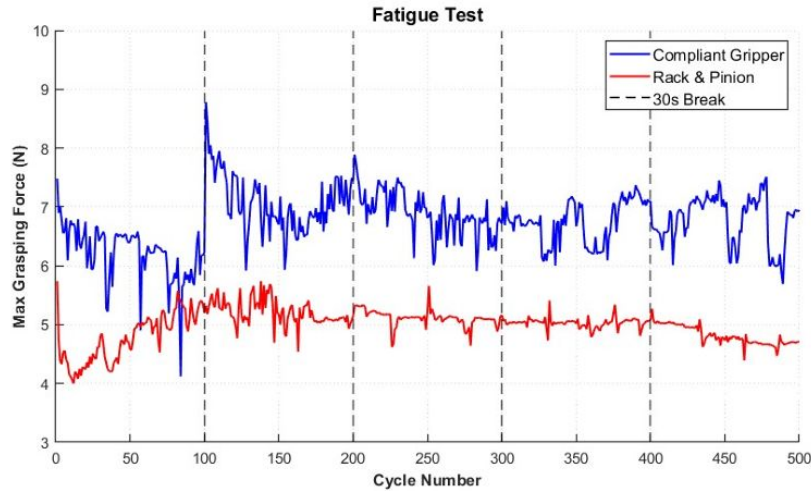
Kinematic Hysteresis (3 trials)



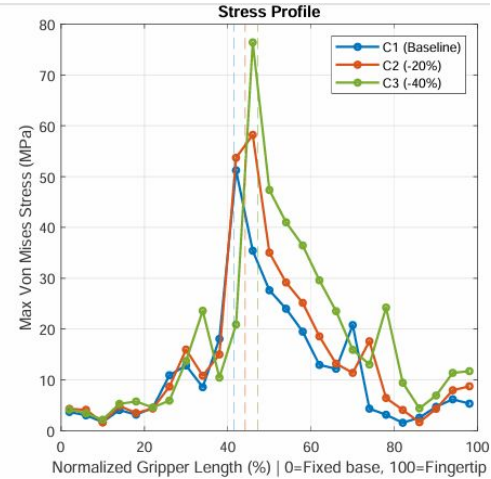
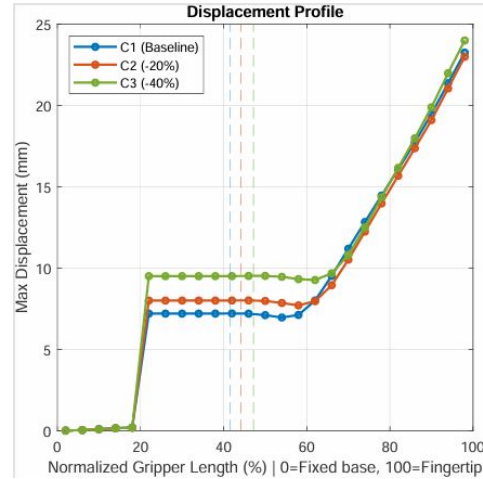
Effective Compliance and Stiffness (3 trials)

Force (N)	Deflection Compliant (mm)	k_C (N/mm)	Defl R&P (mm)	$k_{R\&P}$ (N/mm)
2	0	—	0	—
4	1.31 ± 0.50	1.53 ± 0.58	0.76 ± 0.37	2.63 ± 1.28
6	2.39 ± 0.44	1.67 ± 0.31	1.46 ± 0.48	2.74 ± 0.90
8	3.77 ± 0.30	1.59 ± 0.13	—	—
MAX	7.20 ± 0.64	2.17 ± 0.41	1.89 ± 0.59	2.61 ± 0.83

Appendix

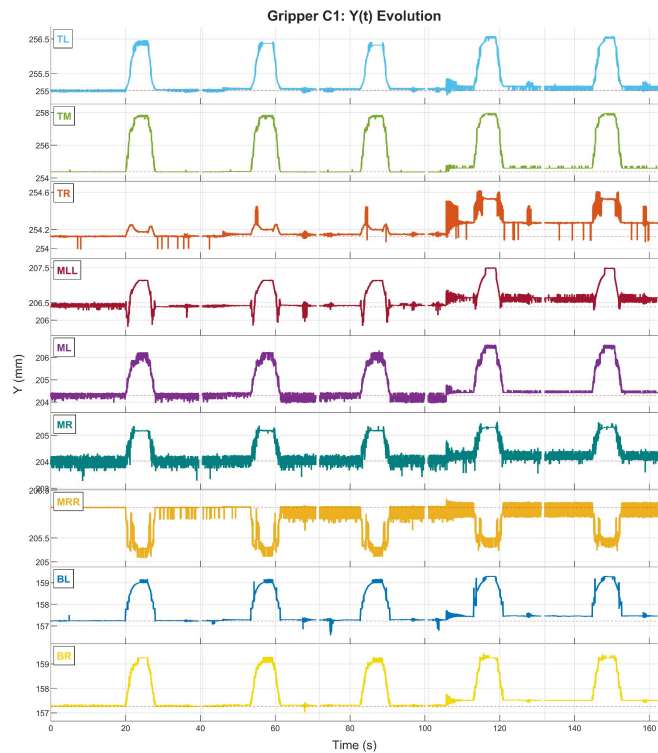


Fatigue Test (500 cycles)



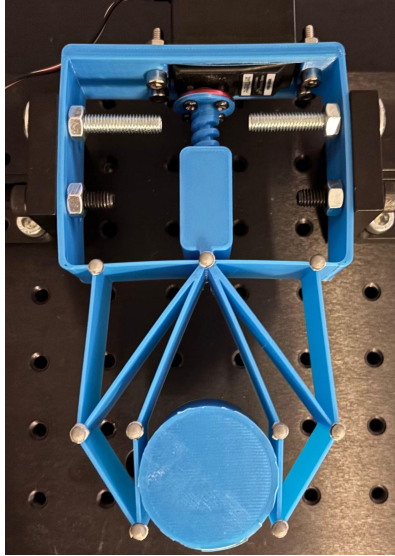
Normalized Displacement and Stress Profiles along the grippers

Appendix

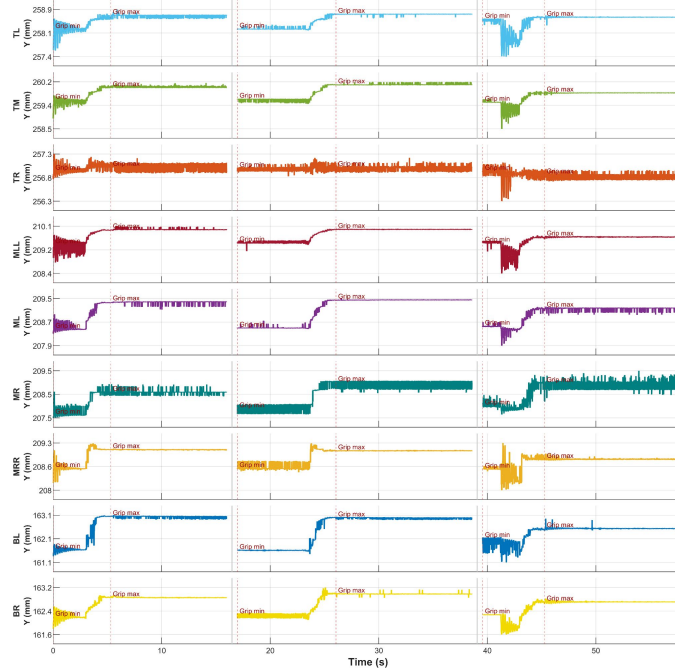


Test A: Y(t) Markers Evolution (5 trials)

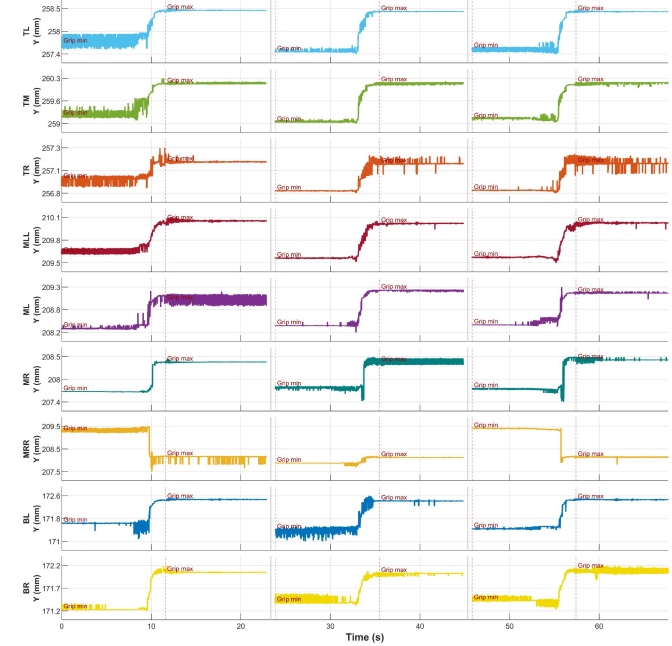
Appendix



Gripper C1 : Y evolution from Grip min to Grip max (3 trials)

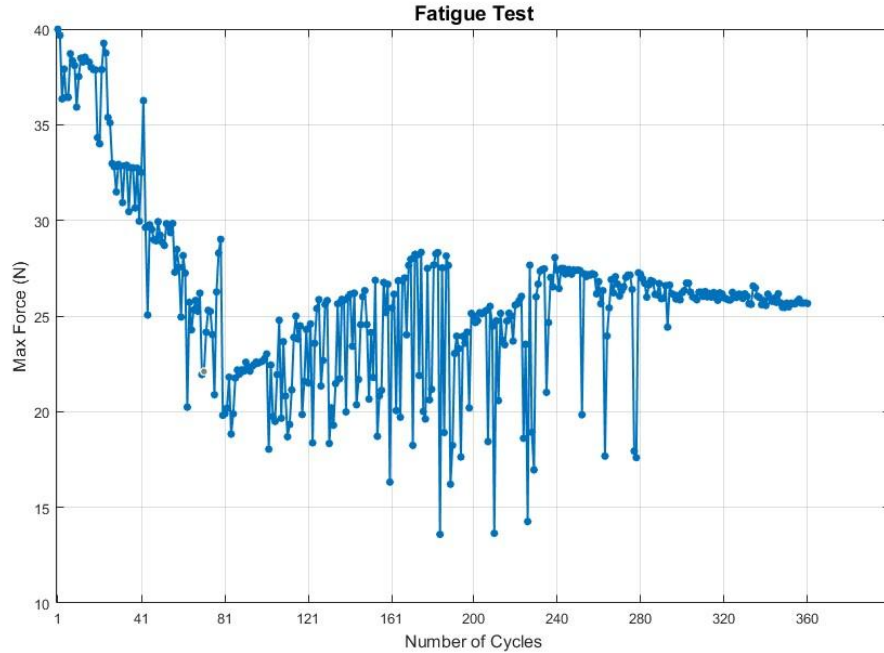


Gripper C2 : Y evolution from Grip min to Grip max (3 trials)

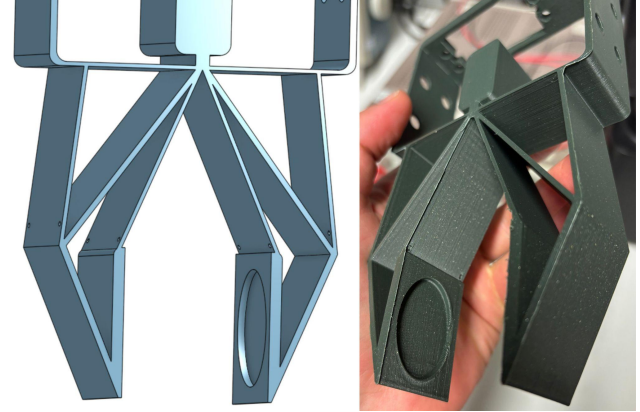


Test B: Y(t) Evolution and Deformation (3 trials)

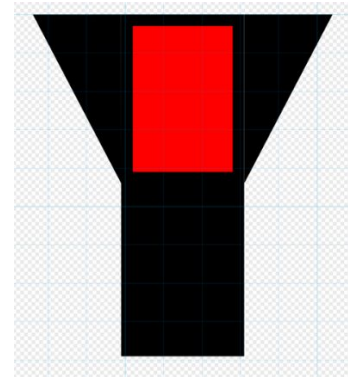
Appendix



Metal Transmission Gripper: Fatigue Test



Design Modification



2D Pattern of the Gloves

Work Completed

- Progress on the report & presentation

2D Design of the gloves

Design for gloves

Design to attach the gripper to the robotic arm

Cut the fabric on Cricut

Tp Description	à faire
Analyse résultat Test 1 + courbes +rapport WEEKLY REPORT : Comparaison avec Caliper pas utile / Faire des recherches sur l'optimisation structure tout avec IA ou autre forme mécanisme compliant / On s'intéresse au compliant mechanism le but est d'évaluer surtout celui là pour voir s'il est mieux qu'un gripper de base / Faire un 2ème holder qui permet de fully constrained le gripper / Faire 2 autres compliant gripper en modifiant 1 paramètre pour comprendre l'influence qu'a ce changement (distance entre bout des doigts et milieu des doigts ou milieu des doigts et base flexible / Refaire les tests où la méthodologie est un peu douteuse et tester les différents compliant sur les tests principaux avec une méthodologie robuste : Pull-out Force / Max Force / Peak Current / Optitrack Etudier la déformation des doigts avec OptiTrack Changer le terme actuation des tableaux Base : 49x51 C2 : 49x40 8 (-20%) Compliant 3 : 49x30.6 (-40%)	The compliant gripper body itself is monolithic, meaning that the flexures and gripping fingers are manufactured as a single continuous part with no assembled joints or hinges. It's also true that the full actuation system is not strictly a single-piece device. So, more accurately, we should describe it as a monolithic compliant gripper with an external screw-based actuation element.) Envoyez mail pour dire les résultats et demander ce que je fais après. Nettoyeur code arduino plus mettre sur github Rapport (état art, photo compliant gripper sans grippera, explication des modifications du dessin pour les tests, expliquer gripper de base. Mettre Fichier CAO sur Github Elargir gripper 0.5cm Enlever bosse holder et extrude 0.15 vers le centre

[illegible]

22

Appendix

 main  1 Branch  0 Tags



 **julien-dubuc** Delete MatLab_Code/Test5 directory 1082f0e · yesterday

 Arduino_Code	Delete Arduino_Code/TestFatigue/s
 Certificates	Delete Certificates/d
 Literature	Delete Literature/s
 MatLab_Code	Delete MatLab_Code/Test5 directory
 Results	Delete Results/s
 Weekly reports	Add files via upload
 README.md	Create README.md

README

Soft-Gripper-Etextiles

Development of an underactuated soft robotic gripper integrated with e-textile tactile sensors. Engineering project at Tyndall National Institute (UCC).

GitHub Deliverables

CR

Collaborative Robotics-MM_Research_Data 

Documents > Supervision > Julien > Docs Handover ▾

All Documents

 Name ▾	Modified ▾	Modified By ▾
 Arduino_Code	5 days ago	Julien Dubuc
 CAD	A few seconds a...	Julien Dubuc
 Literature	5 days ago	Julien Dubuc
 MatLab_Code	5 days ago	Julien Dubuc
 Results	5 days ago	Julien Dubuc
 Simscale	5 days ago	Julien Dubuc
 Weekly reports	5 days ago	Julien Dubuc

OneDrive Deliverables